A multimodal spatial-attention multiple-instance learning method for cervical cancer recurrence prediction on pretreatment planning CT and PET imaging data

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Introduction: Cervical cancer recurrence prediction is critical for treatment planning and patient outcomes. While multimodal planning CT and PET improve predictive power over single-modality models, they introduce technical challenges. Multimodal datasets often exhibit site-specific acquisition variability, subtle prognostic cues, and missing modalities when either PET or CT is unavailable. To address these limitations, multimodal AI approaches, particularly Multiple-Instance Learning (MIL), enable robust local evidence aggregation. In this work, we propose a tumor-neighborhood spatial-attention MIL approach that improves outcome prediction by processing overlapping subvolumes from the tumor and adjacent tissue. This design reduces sensitivity to nuisance variation and modality incompleteness while preserving complementary PET/CT information. Furthermore, learned attention weights provide interpretability by identifying tumor-neighborhood regions that most strongly drive patient-level recurrence prediction.

Method: We propose a tumor-neighborhood spatial-attention MIL model that characterizes the the tumor and its surrounding environment by extracting multiple 3D subvolumes from planning CT and PET (Fig. 1). The architecture consists of (1) dual modality-specific Deep Layer Aggregation 3D (DLA3D) encoders that learn local representations from CT and PET subvolumes, and (2) a gated-attention MIL pooling module that aggregates evidence across tumor-neighborhood regions to form a global feature for patient-level prediction. This modular design supports flexible fusion and remains robust when one modality is missing.

Furthermore, the gated-attention mechanism enables better model interpretability by assigning an importance score to each tumor-neighborhood subvolume, reflecting its contribution to the final recurrence prediction. These weights are used to identify informative regions within the multimodal space. We validate the attention-derived importance via an ablation study by removing the top-ranked subvolumes (Top-3) at inference and re-evaluating performance; the resulting drop in predicted probability quantifies the reliance on the removed evidence (Fig. 2).

Results: In our work, we frame recurrence prediction as a binary classification task (recurrent vs non-recurrent). We included 278 patients with paired pretreatment PET/CT scans and evaluated the model using five-fold cross-validation with an 80%/20% train/validation split per fold. Results show that mean AUC improved from 0.672 to 0.713 (+0.041), and mean accuracy improved from 0.572 to 0.666 (+0.094). Specificity increased from 0.485 to 0.696 (+0.211), while sensitivity decreased from 0.715 to 0.627 (-0.088). Mean FP decreased from 15.2 to 9.2 (-6.0), and mean FN increased from 4.8 to 6.2 (+1.4) (Table 1). For interpretability, we performed an ablation-based Top-3 patch removal test using attention-based evidence ranking. On two representative folds, the highest-AUC fold showed marginal significance (Mann–Whitney $p = 0.051$), while the fold with the largest overall gain was significant ($p < 0.05$) (Fig. 2). In both cases, a larger confidence drop after removing top-ranked patches supports that attention localizes prognostic tumor-neighborhood cues.

Conclusions: Our tumor-neighborhood spatial-attention MIL framework improves outcome prediction robustness over a registered planning CT+PET baseline by aggregating multiple overlapping subvolumes, while providing instance-level interpretability via attention-based patch ranking supported by top-patch removal.

Potential impacts: This framework may enable more robust pelvic PET/CT outcome prediction and supports future directions including cross-site generalization, controlled fusion of additional modalities, and patch-level interpretability through evidence ranking.

Supplementary Page

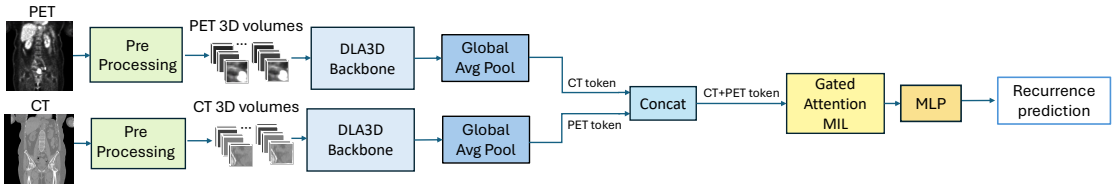

Figure 1: Framework of the proposed tumor-neighborhood multi-view MIL model. Multiple tumor-neighborhood 3D subvolumes from registered pretreatment CT and PET are encoded by modality-specific DLA3D backbones with global average pooling. The resulting tokens are concatenated and aggregated via gated-attention MIL pooling, and an MLP outputs the patient-level recurrence prediction.

	AUC		TP		TN		FP		FN		Sensitivity		Specificity		Accuracy	
Fold	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/
1	0.739	**0.772**	10	9	20	19	11	12	2	3	**0.833**	0.750	**0.645**	0.613	**0.698**	0.651
2	0.666	**0.763**	17	13	8	20	18	6	2	6	**0.895**	0.684	0.308	**0.769**	0.556	**0.733**
3	**0.734**	0.704	10	5	17	22	10	5	6	11	**0.625**	0.313	0.630	**0.815**	**0.628**	0.628
4	0.595	**0.658**	10	12	13	22	18	9	8	6	0.556	**0.667**	0.419	**0.710**	0.469	**0.694**
5	0.626	**0.670**	12	13	14	19	19	14	6	5	0.667	**0.722**	0.424	**0.576**	0.510	**0.627**
Mean	0.672	**0.713**	–		–		–		–		**0.715**	0.627	0.485	**0.696**	0.572	**0.666**

Table 1: Five-fold performance comparison between models trained without and with MIL. For AUC, sensitivity, specificity, and accuracy, the better value within each fold is highlighted in bold; the better mean value is highlighted in blue. Confusion-matrix counts are omitted for the mean row.

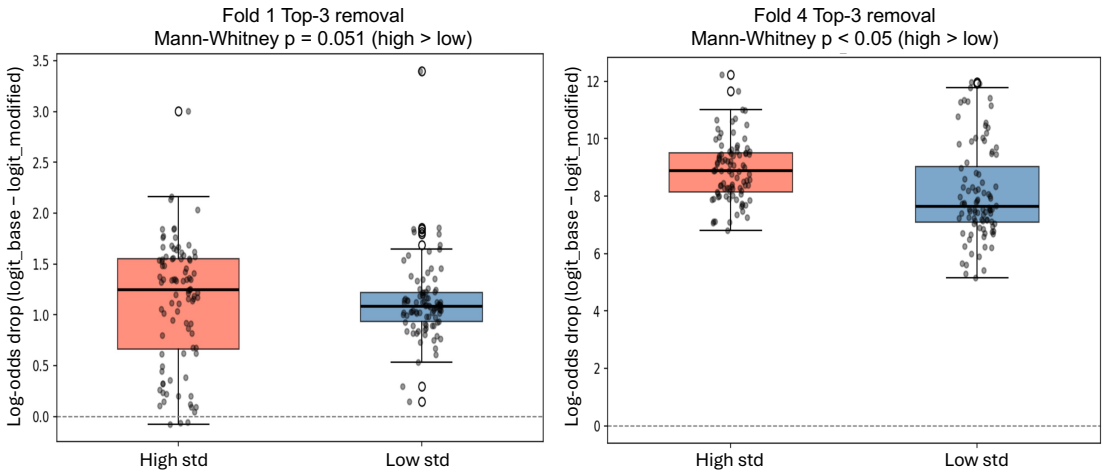

Figure 2: Ablation-based interpretability via Top-3 patch removal. Left: the fold achieving the highest AUC (Mann–Whitney $p = 0.051$). Right: the fold with the largest overall metric gain (Mann–Whitney $p < 0.001$). For each patient, we rank tumor-neighborhood subvolumes by MIL attention weights and remove the Top-3 ranked patches at inference. The y-axis shows the log-odds drop, $\Delta = \text{logit}(p_{\text{base}}) - \text{logit}(p_{\text{removed}})$, measuring the decrease in predicted probability after removal. The x-axis compares cases whose removed Top-3 patches have high versus low within-patch intensity standard deviation (STD), defined by tertiles across patients (top vs bottom tertile).